

Statistics Homework 6: Multiple Comparisons

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Part A: Racial/Ethnic group differences in couples

The results below are from a dataset used in a study by Sanford (2007). All the participants in this study were married and they all completed a measure of relationship satisfaction called the Quality Marriage Index (QMI). The sample was divided into four racial/ethnic categories (titled White, Black, Hispanic, and Asian), and ANOVA was used to investigate differences between the four groups in their reported levels of relationship satisfaction. Use the results printed below to reconstruct the ANOVA table.

Descriptives

qmi

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
white	45	4.62	1.40	.21	4.20	5.04	1.60	6.00
black	41	4.50	1.27	.20	4.10	4.90	1.80	6.00
hispanic	42	5.16	1.06	.16	4.83	5.49	1.00	6.00
asian	40	4.75	1.09	.17	4.41	5.10	2.00	6.00
Total	168	4.76	1.23	.10	4.57	4.94	1.00	6.00

ANOVA

qmi

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups					
Within Groups					
Total					

1. Compute the total sum of squares.
2. Compute the sum of squares between groups.
3. Compute the sum of squares within groups.
4. Compute the degrees of freedom between groups.
5. Compute degrees of freedom within groups.
6. Compute total degrees of freedom.
7. Compute mean square between groups.
8. Compute mean square within groups.

Is there a difference between groups? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

9. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
10. the value of the descriptive statistic
11. the name of the inferential statistic (use the Acceptable Abbreviations Page)
12. the value of the inferential statistic
13. the p-value
14. the name of a descriptive statistic that has been corrected for bias
15. the value of a descriptive statistic that has been corrected for bias

16. Was there a significant difference between the four groups?

(A) yes (B) no

Part B: Different ways to compare two means

Consider the following descriptive statistics for two groups:

Group Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Group 1 (treatment)	62	2.98	.80	.10
Group 0 (control)	64	3.46	1.06	.13

A *t*-test was used to determine if group1 scored lower than group 2, and there was a significant difference between groups ($t = 2.86, p = .005$).

Based on the information above:

17. What is the *d*-statistic?

18. What is the *p* value for the *d* statistic?

19. What is $\frac{\text{COV}_{xy}}{sd_x sd_y}$?

20. What is the *p*-value for the correlation?

21. What is $\frac{\bar{x}_1 - \bar{x}_2}{sd_{\bar{x}_1 - \bar{x}_2}}$?

22. What is $\frac{r}{sd_r}$?

Now, assume you analyze the above data using an ANOVA to test for differences between means. Based on the information above:

23. What is eta?

24. What is eta squared?

25. What is the *p*-value for eta squared?

26. What is the predicted value for someone in group 1?

Now, assume you that you analyze the above data using regression and assume you assign scores to people using dummy codes such that everyone in Group 1 (the treatment group) receives a score of 1 and everyone in Group 0 (the control group) receives a score of 0.

27. What is the intercept?

28. What is $\frac{\text{COV}_{xy}}{sd_x^2}$?

29. What is $\frac{b}{sd_b}$?
30. What is R^2 ?
31. What is the p -value for R^2 ?
32. What is the standardized beta weight?
33. What is the p -value for the standardized beta weight?
34. What is the value for $\hat{y}$ for someone in group 1?

Part C: Orthogonal contrasts

Orthogonal contrasts are being used to create contrasts for an analysis with three groups. The code assignment chart depicted below has four missing values (labeled E, F, G, H).

x_1	E	.33	F
x_2	-.5	G	H

35. Following the procedures discussed in class, what does the value for E need to be?
36. what does the value for F need to be?
37. what does the value for G need to be?
38. what does the value for H need to be?

Depicted below is a code assignment chart and results from an analysis comparing three groups:

	Group E	Group F	Group G
x_1	-.66	.33	.33
x_2	0	-.5	.5

$$\hat{y} = b_0 + 5x_1 + 4x_2 \quad \bar{y}_{\text{Group F}} = 10$$

39. Based in the information provided above, what is the mean for Group G?
40. What is the mean for group E?
41. What is b_0 ?

Depicted below is a code assignment chart and group means for three groups:

	Group E $\bar{x} = 10$	Group F $\bar{x} = 12$	Group G $\bar{x} = 18$
x_1	-.66	.33	.33
x_2	0	-.5	.5

Assume you put these contrasts into the following equation:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

42. Based in the information provided above, what is b_0 ?
43. What is b_1 ?
44. What is b_2 ?

Describe the process (as discussed in class) for creating orthogonal contrasts. Fill in each of the following blanks with a numerical value:

45. For each contrast, the sum of the coded values need to be equal to _____.
46. The results are easiest to interpret if the non-zero codes are exactly _____ unit(s) apart from each other.
47. Each contrast needs to contain _____ different non-zero values.
48. If you do not want a group to be included in a contrast, that group should be assigned a code of _____.

Part D: Cohabitation and relationship satisfaction

Studies sometimes find lower levels of relationship satisfaction in married couples that cohabited before getting married and higher satisfaction in those that did not cohabit before getting married. Kline and her colleagues (2004) wondered if there is a difference between couples that cohabit before they get engaged versus couples that cohabit only after getting engaged. They conducted a study with 133 newly married couples and both partners of each couple completed a measure of relationship satisfaction. The sample was divided into four groups. Group 1 cohabited prior to getting engaged. Group 2 cohabited after getting engaged and before they got married. Group 3 did not live together prior to getting married. Use the results printed below to reconstruct the ANOVA table testing for group differences in relationship satisfaction.

Descriptives

quality

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1	58	120.67	12.92	1.70	117.28	124.07	78.15	141.13
2	25	129.23	14.93	2.99	123.07	135.40	104.44	160.88
3	50	132.76	14.64	2.07	128.60	136.92	99.88	171.08
Total	133	126.83	14.94	1.30	124.26	129.39	78.15	171.08

ANOVA

quality

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups					
Within Groups					
Total					

49. Compute the total sum of squares.
50. Compute the sum of squares between groups.
51. Compute the sum of squares within groups.
52. Compute the degrees of freedom between groups.
53. Compute degrees of freedom within groups.
54. Compute total degrees of freedom.
55. Compute mean square between groups.
56. Compute mean square within groups.

Is there a difference between groups? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

57. the name of the descriptive statistic (use the Acceptable Abbreviations Page)

58. the value of the descriptive statistic

59. the name of the inferential statistic (use the Acceptable Abbreviations Page)

60. the value of the inferential statistic

61. the p-value

62. the name of a descriptive statistic that has been corrected for bias

63. the value of a descriptive statistic that has been corrected for bias

64. Was there a significant difference between the four groups?

(A) yes (B) no

Part E: Cohabitation and conflict communication skill

The previous section reported results from Kline's (2004) study comparing three groups of married couples in their levels of relationship satisfaction. Kline also investigated differences between the three groups in their use of negative communication during conflict conversations. Each couple was video recorded as they attempted to resolve an area of relationship conflict, and a team of research assistants rated the level of negative communication (which was defined to include withdrawal, hostility, and negative emotion).

Results for negative communication are printed below. For these results, do NOT make all the possible comparisons, and instead, focus on two *a-priori* hypotheses. The **first hypothesis** is that the couples cohabiting before engagement (Group 1) will use more negative communication than the groups waiting to cohabit until after some form of commitment to marriage had been made (Groups 2 and 3). The **second hypothesis** is that couples cohabiting after engagement (Group 2) will use more negative communication than couples cohabiting after marriage (Group 3).

Descriptives

observed_neg

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1	58	3.62	1.39		3.25	3.98	-.60	6.34
2	25	2.96	1.42	.28	2.37	3.55	.48	6.77
3	50	2.91	1.17	.17	2.58	3.24	.70	6.27
Total	133	3.23		.12	3.00	3.46	-.60	6.77

ANOVA

observed_neg

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	15.72	2	7.86	4.52	.013
Within Groups	226.04	130	1.74		
Total	241.75	132			

Contrast Coefficients

Contrast	group		
	1	2	3
1	-.66	.33	.33
2	0	-.5	.5

Contrast Tests

			Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
observed_neg	Assume equal variances	Contrast 1	-.45	.16	-2.89	130	.005
		2	-.02	.16	-.15	130	.878
	Does not assume equal variances	1	-.45	.16	-2.78	97.35	.006
		2	-.02	.16	-.15	40.76	.881

Multiple Comparisons

Dependent Variable: observed_neg

			Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
(I) group	(J) group	Lower Bound				Upper Bound	
Tukey HSD	1	2	.659	.315	.096	-.09	1.41
		3	.709 [*]	.254	.017	.11	1.31
	2	1	-.659	.315	.096	-1.41	.09
		3	.050	.323	.987	-.72	.82
	3	1	-.709 [*]	.254	.017	-1.31	-.11
		2	-.050	.323	.987	-.82	.72
LSD	1	2	.659 [*]	.315	.039	.04	1.28
		3	.709 [*]	.254	.006	.21	1.21
	2	1	-.659 [*]	.315	.039	-1.28	-.04
		3	.050	.323	.878	-.59	.69
	3	1	-.709 [*]	.254	.006	-1.21	-.21
		2	-.050	.323	.878	-.69	.59
Bonferroni	1	2	.659	.315	.116	-.11	1.42
		3	.709 [*]	.254	.018	.09	1.33
	2	1	-.659	.315	.116	-1.42	.11
		3	.050	.323	1.000	-.73	.83
	3	1	-.709 [*]	.254	.018	-1.33	-.09
		2	-.050	.323	1.000	-.83	.73

*. The mean difference is significant at the 0.05 level.

65. What is the standard error of the mean for group 1?

66. What is $\sum(y - \bar{y})^2$?67. What is $\frac{\sum(y - \bar{y})^2}{n-1}$?

68. What is the average distance that people's scores on the outcome variable deviated from the grand mean?
69. What is the average distance that people's scores on the outcome variable deviated from their individual group means? (Calculate a value analogous to the standard error of estimate.)
70. What is the percent of variance that has been explained in this equation?
71. What is the expected score for y for a couple that is in group 3?
72. Assume the Jones are a married couple participating in the study. If the Jones are in Group 1, what is the Jones' predicted score on the outcome variable?
73. If the Jones actually received a score of 2 on the outcome variable, what is the squared error for the Jones' score?
74. According to the results printed above, what is the typical squared error for participants in this study?
75. Was the squared error for the Jones higher or lower than the typical squared error for participants in this study?
76. According to the results printed above, what is the sum of squared errors for all participants in this study?
77. If the two contrasts are used to predict the outcome variable in a regression equation, what is b_0 ?
78. What is b_1 ?
79. What is b_2 ?

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p -value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Is there a difference between groups in their use of negative communication?

- | | |
|---|--|
| 80. What is the name of the descriptive statistic? | 82. What is the name of the inferential statistic? |
| 81. What is the value of the descriptive statistic? | 83. What is the value of the inferential statistic? |
| | 84. What is the p value (exactly as printed)? |

Do the data support hypothesis 1?

- | | |
|---|--|
| 85. What is the name of the descriptive statistic? | 87. What is the name of the inferential statistic? |
| 86. What is the value of the descriptive statistic? | 88. What is the value of the inferential statistic? |
| | 89. What is the p value (exactly as printed)? |

Do the data support hypothesis 2?

- | | |
|---|--|
| 90. What is the name of the descriptive statistic? | 92. What is the name of the inferential statistic? |
| 91. What is the value of the descriptive statistic? | 93. What is the value of the inferential statistic? |
| | 94. What is the p value (exactly as printed)? |

95. Consider the results pertaining to hypothesis 1. What best describes the size of this effect?

- (A) not significant (B) significant but trivial (C) small (D) medium (E) large

96. Consider the results pertaining to hypothesis 2. What best describes the size of this effect?

- (A) not significant (B) significant but trivial (C) small (D) medium (E) large

97. In a previous question, you were asked to give a p -value for hypothesis 1. What would the p -value be if you used the Bonferroni correction?

Part F: Online stress management intervention and depression

Hintz and his colleagues (2015) devised an online stress management intervention for college students. The intervention consisted of four modules in which participants first watched a video of an expert discussing a topic related to stress management, and then they completed an application exercise, responded to some motivational questions, and received instruction on keeping a stress log. Hintz tested this intervention using a sample of 204 college students all reporting at least moderate levels of current stress, and participants were randomly assigned into three groups. Group 1 received the stress management intervention. Group 2 received the intervention plus automated feedback after completing each module. Group 3 was the control group and they did not receive the intervention. After the intervention, all participants completed a measure of depression.

Results are printed below. For these results, do NOT make all the possible comparisons, and instead, focus on two *a-priori* hypotheses. The first hypothesis is that the two treatment groups (Group 1 and Group 2) will report less depression than the control group. The second hypothesis is that the treatment group receiving feedback (Group 2) will report less depression than the treatment group receiving the intervention without feedback (Group 1).

Descriptives

Depression

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1.00	60	3.99	4.02	.52	2.95	5.03	-3.18	13.56
2.00	76	4.55	4.15	.48	3.60	5.50	-5.00	16.10
3.00	68	5.95	4.94	.60	4.75	7.15	-4.04	19.35
Total	204	4.85	4.45	.31	4.24	5.47	-5.00	19.35

ANOVA

Depression

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	133.49	2	66.75	3.46	.0334
Within Groups	3880.00	201	19.30		
Total	4013.49	203			

Contrast Coefficients

Contrast	group		
	1.00	2.00	3.00
1	.33	.33	-.66
2	-.5	.5	0

Contrast Tests

			Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
Depression	Assume equal variances	1	-1.11	.43	-2.57	201	.0109
		2	.28	.38	.74	201	.4614
	Does not assume equal variances	1	-1.11	.46	-2.42	114.18	.0172
		2	.28	.35	.80	128.57	.4279

Multiple Comparisons

Dependent Variable: Depression

			Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
(I) group	(J) group					Lower Bound	Upper Bound
Tukey HSD	1.00	2.00	-.56	.76	.7412	-2.35	1.23
		3.00	-1.96*	.78	.0335	-3.80	-.12
	2.00	1.00	.56	.76	.7412	-1.23	2.35
		3.00	-1.40	.73	.1388	-3.13	.33
	3.00	1.00	1.96*	.78	.0335	.12	3.80
		2.00	1.40	.73	.1388	-.33	3.13
LSD	1.00	2.00	-.56	.76	.4614	-2.06	.94
		3.00	-1.96*	.78	.0126	-3.49	-.43
	2.00	1.00	.56	.76	.4614	-.94	2.06
		3.00	-1.40	.73	.0577	-2.85	.05
	3.00	1.00	1.96*	.78	.0126	.43	3.49
		2.00	1.40	.73	.0577	-.05	2.85
Bonferroni	1.00	2.00	-.56	.76	1.0000	-2.39	1.27
		3.00	-1.96*	.78	.0377	-3.84	-.08
	2.00	1.00	.56	.76	1.0000	-1.27	2.39
		3.00	-1.40	.73	.1731	-3.17	.37
	3.00	1.00	1.96*	.78	.0377	.08	3.84
		2.00	1.40	.73	.1731	-.37	3.17

*. The mean difference is significant at the 0.05 level.

98. If the two contrasts are used to predict the outcome variable in a regression equation, what is b_0 ?99. What is b_1 ?100. What is b_2 ?

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Is there a difference between groups in their levels of depression?

101. What is the name of the descriptive statistic?

102. What is the value of the descriptive statistic?

103. What is the name of the inferential statistic?

104. What is the value of the inferential statistic?

105. What is the p value (**exactly as printed**)?

Do the data support hypothesis 1?

106. What is the name of the descriptive statistic? 108. What is the name of the inferential statistic?
 107. What is the value of the descriptive statistic? 109. What is the value of the inferential statistic?
 110. What is the p value (**exactly as printed**)?

Do the data support hypothesis 2?

111. What is the name of the descriptive statistic? 113. What is the name of the inferential statistic?
 112. What is the value of the descriptive statistic? 114. What is the value of the inferential statistic?
 115. What is the p value (**exactly as printed**)?

116. Consider the results pertaining to hypothesis 1. What best describes the size of this effect?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

117. Consider the results pertaining to hypothesis 2. What best describes the size of this effect ?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

118. In a previous question, you were asked to give a p-value for hypothesis 1. What would the p-value be if you used the Bonferroni correction?

119. In a previous question, you were asked to give a p-value for hypothesis 2. What would the p-value be if you used the Bonferroni correction?

Part G: Online stress management intervention and perceived stress

The previous section reported results from Hintz's (2015) study comparing three groups of college students in their levels of depression. Hintz also measured students' levels of perceived stress and these results are printed below. For these results, take a **different** approach to analysis than was used in the previous section. Do NOT use planned comparisons, and instead, **make all the possible comparisons**.

Descriptives

Stress

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1.00	60	1.76	.66		1.59	1.93	.21	2.98
2.00	76	1.80	.58	.07	1.67	1.93	.70	3.43
3.00	68	2.04	.64	.08	1.89	2.20	.12	3.45
Total	204	1.87		.04	1.78	1.96	.12	3.45

ANOVA

Stress

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	3.15	2	1.57	4.04	.019
Within Groups	78.24	201	.39		
Total	81.39	203			

Contrast Coefficients

Contrast	group		
	1.00	2.00	3.00
1	.33	.33	-.66
2	-.5	.5	0

Contrast Tests

			Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
Contrast							
Stress	Assume equal variances	1	-.17	.06	-2.84	201	.005
		2	.02	.05	.37	201	.710
	Does not assume equal variances	1	-.17	.06	-2.79	130.36	.006
		2	.02	.05	.37	117.55	.712

Multiple Comparisons

Dependent Variable: Stress

			Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
	(I) group	(J) group					
Tukey HSD	1.00	2.00	-.04	.11	.927	-.29	.21
		3.00	-.28 [*]	.11	.030	-.54	-.02
	2.00	1.00	.04	.11	.927	-.21	.29
		3.00	-.24	.10	.053	-.49	.00
	3.00	1.00	.28 [*]	.11	.030	.02	.54
		2.00	.24	.10	.053	.00	.49
LSD	1.00	2.00	-.04	.11	.710	-.25	.17
		3.00	-.28 [*]	.11	.011	-.50	-.07
	2.00	1.00	.04	.11	.710	-.17	.25
		3.00	-.24 [*]	.10	.020	-.45	-.04
	3.00	1.00	.28 [*]	.11	.011	.07	.50
		2.00	.24 [*]	.10	.020	.04	.45
Bonferroni	1.00	2.00	-.04	.11		-.30	.22
		3.00	-.28 [*]	.11		-.55	-.02
	2.00	1.00	.04	.11		-.22	.30
		3.00	-.24	.10		-.49	.01
	3.00	1.00	.28 [*]	.11		.02	.55
		2.00	.24	.10		-.01	.49

*. The mean difference is significant at the 0.05 level.

120. What is the standard error of the mean for group 1?
121. What is $\Sigma(y - \bar{y})^2$?
122. What is $\frac{\Sigma(y - \bar{y})^2}{n - 1}$?
123. What is the average distance that people's scores on the outcome variable deviated from the grand mean?
124. What is the average distance that people's scores on the outcome variable deviated from their individual group means? (Calculate a value analogous to the standard error of estimate.)
125. What is the percent of variance that has been explained in this equation?
126. What is the expected score for y for a person that is in group 3?
127. If Lucy is Group 1, what is Lucy's predicted score on the outcome variable?
128. If Lucy actually received a score of 2.16 on the outcome variable, what is the squared error for Lucy's score?
129. According to the results printed above, what is the typical squared error for participants in this study?
130. Was the squared error for Lucy higher or lower than the typical squared error for participants in this study?
131. According to the results printed above, what is the sum of squared errors for all participants in this study?

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Is there a difference between groups in levels of perceived stress?

- | | |
|--|---|
| 132. What is the name of the descriptive statistic? | 134. What is the name of the inferential statistic? |
| 133. What is the value of the descriptive statistic? | 135. What is the value of the inferential statistic? |
| | 136. What is the p value (exactly as printed)? |

Is group 1 different from group 2?

- | | |
|--|---|
| 137. What is the name of the descriptive statistic? | 139. What is the name of the inferential statistic? |
| 138. What is the value of the descriptive statistic? | 140. What is the value of the inferential statistic? |
| | 141. What is the p value (exactly as printed)? |

Is group 1 different from group 3?

- | | |
|--|---|
| 142. What is the name of the descriptive statistic? | 144. What is the name of the inferential statistic? |
| 143. What is the value of the descriptive statistic? | 145. What is the value of the inferential statistic? |
| | 146. What is the p value (exactly as printed)? |

Is group 2 different from group 3?

- | | |
|--|---|
| 147. What is the name of the descriptive statistic? | 149. What is the name of the inferential statistic? |
| 148. What is the value of the descriptive statistic? | 150. What is the value of the inferential statistic? |
| | 151. What is the p value (exactly as printed)? |

152. Consider the comparison between group 1 and group 2. What best describes the size of this effect ?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

153. Consider the comparison between group 1 and group 3. What best describes the size of this effect ?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large
154. Consider the comparison between group 2 and group 3. What best describes the size of this effect ?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large
155. In a previous question, you were asked to give a p-value for the comparison between group 1 and **group 3**. What would the p-value be if you used the Bonferroni correction?
156. In a previous question, you were asked to give a p-value for the comparison between group 2 and group 3. What would the p-value be if you used the Bonferroni correction?
157. Consider the comparison between group 1 and group 2. What would the t -value be if you used Fisher's LSD to test for group differences?
158. Consider the comparison between group 1 and group 3. What would the t -value be if you used Fisher's LSD to test for group differences?
159. Consider the comparison between group 2 and group 3. What would the t -value be if you used Fisher's LSD to test for group differences?

Part H: Rats in a water maze

Liu and colleagues (2015) conducted a study with rats to investigate the extent to which the drug hyperforin relieves memory impairments associated with living in a stressful environment. Liu used a sample of 30 rats which were assigned to three conditions. Group 1 was a control condition in which rats were reared following normal laboratory procedures. Group 2 was a stress condition in which rats were subjected to three weeks of chronic, unpredictable stressors. Group 3 was a stress plus hyperforin condition in which rats were subjected to the same stressors, but also received the drug hyperforin. Memory was assessed using a water maze. The water maze is a pool of water with a hidden platform, and rats are placed in the pool and given repeated trials so they can learn how to swim to the hidden platform. The outcome variable for this study is the number of seconds that it takes a rat to swim to the platform after having an opportunity to learn the maze.

Results (estimated from the visual figures published by Liu) are printed below. For this analysis, do NOT use planned comparisons, and instead, **make all the possible comparisons**.

Descriptives

Seconds

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1.00	12	37.00	8.33	2.41	31.71	42.29	26.65	55.46
2.00	10	49.00	9.56	3.02	42.16	55.84	30.33	68.72
3.00	8	42.00	7.90	2.79	35.40	48.60	29.72	51.91
Total	30	42.33	9.84	1.80	38.66	46.01	26.65	68.72

ANOVA

Seconds

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	786.67	2	393.34	5.25	.012
Within Groups	2022.35	27	74.90		
Total	2809.02	29			

Contrast Coefficients

Contrast	group		
	1.00	2.00	3.00
1	-.66	.33	.33
2	0	-.5	.5

Contrast Tests

			Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
Seconds	Assume equal variances	Contrast 1	5.61	2.13	2.63	27	.014
		2	-3.50	2.05	-1.71	27	.100
	Does not assume equal variances	1	5.61	2.09	2.69	24.09	.013
		2	-3.50	2.06	-1.70	15.97	.108

Multiple Comparisons

Dependent Variable: Seconds

			Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
	(I) group	(J) group					
Tukey HSD	1.00	2.00	-12.00 [*]	3.71	.009	-21.19	-2.81
		3.00	-5.00	3.95	.426	-14.79	4.79
	2.00	1.00	12.00 [*]	3.71	.009	2.81	21.19
		3.00	7.00	4.11	.222	-3.18	17.18
	3.00	1.00	5.00	3.95	.426	-4.79	14.79
		2.00	-7.00	4.11	.222	-17.18	3.18
LSD	1.00	2.00	-12.00 [*]	3.71	.003	-19.60	-4.40
		3.00	-5.00	3.95	.216	-13.11	3.11
	2.00	1.00	12.00 [*]	3.71	.003	4.40	19.60
		3.00	7.00	4.11	.100	-1.42	15.42
	3.00	1.00	5.00	3.95	.216	-3.11	13.11
		2.00	-7.00	4.11	.100	-15.42	1.42
Bonferroni	1.00	2.00	-12.00 [*]	3.71		-21.46	-2.54
		3.00	-5.00	3.95		-15.08	5.08
	2.00	1.00	12.00 [*]	3.71		2.54	21.46
		3.00	7.00	4.11		-3.48	17.48
	3.00	1.00	5.00	3.95		-5.08	15.08
		2.00	-7.00	4.11		-17.48	3.48

*. The mean difference is significant at the 0.05 level.

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p -value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Is there a difference between rat groups in the time it took to find the hidden platform?

- | | |
|--|---|
| 160. What is the name of the descriptive statistic? | 162. What is the name of the inferential statistic? |
| 161. What is the value of the descriptive statistic? | 163. What is the value of the inferential statistic? |
| | 164. What is the p value (exactly as printed)? |

Is group 1 different from group 2?

- | | |
|--|---|
| 165. What is the name of the descriptive statistic? | 167. What is the name of the inferential statistic? |
| 166. What is the value of the descriptive statistic? | 168. What is the value of the inferential statistic? |
| | 169. What is the p value (exactly as printed)? |

Is group 1 different from group 3?

- | | |
|--|---|
| 170. What is the name of the descriptive statistic? | 172. What is the name of the inferential statistic? |
| 171. What is the value of the descriptive statistic? | 173. What is the value of the inferential statistic? |
| | 174. What is the p value (exactly as printed)? |

Is group 2 different from group 3?

- | | |
|--|---|
| 175. What is the name of the descriptive statistic? | 177. What is the name of the inferential statistic? |
| 176. What is the value of the descriptive statistic? | 178. What is the value of the inferential statistic? |
| | 179. What is the p value (exactly as printed)? |

180. Consider the comparison between group 1 and group 2. What best describes the size of this effect?
(A) not significant (B) significant but trivial (C) small (D) medium (E) large

181. Consider the comparison between group 1 and group 3. What best describes the size of this effect?
(A) not significant (B) significant but trivial (C) small (D) medium (E) large

182. Consider the comparison between group 2 and group 3. What best describes the size of this effect?
(A) not significant (B) significant but trivial (C) small (D) medium (E) large

183. In a previous question, you were asked to give a p -value for the comparison between group 1 and group 2. What would the p -value be if you used the Bonferroni correction?

184. In a previous question, you were asked to give a p -value for the comparison between group 1 and group 3. What would the p -value be if you used the Bonferroni correction?

185. Consider the comparison between group 1 and group 2. What would the t -value be if you used Fisher's LSD to test for group differences?

186. Consider the comparison between group 1 and group 3. What would the t -value be if you used Fisher's LSD to test for group differences?

187. Consider the comparison between group 2 and group 3. What would the t -value be if you used Fisher's LSD to test for group differences?

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